

Subject Description Form

Subject Code	AMA571
Subject Title	Financial Technology
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To introduce students some fundamental building blocks in blockchain technology as well as its application in cryptocurrencies, stablecoins, and decentralized finance. Help students to develop practical programming, statistics and mathematical skills that are valued in the FinTech industry.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Acquire knowledge of the core ideas in FinTech as well as develop hands-on programming skills in the Fintech area. (b) Analyze financial problems, and identify and define the computing technologies appropriate to its solution. (c) Gain experience in designing and implementing smart contracts and decentralized applications. (d) Understand the blockchain technology and cryptocurrencies. (e) Reflect the quantitative skills to solve other financial applications.
Subject Synopsis/ Indicative Syllabus	<u>Solidity for smart contract</u> <u>Reinforcement learning in investment</u> <u>Blockchain Technology</u> Cryptographic hash function, public-key cryptography and digital signature, block building, mining and PoW <u>Smart Contract</u> Basics, solidity essentials, solidity dive <u>Cryptocurrency</u> Transactions and wallets, protocols <u>Decentralized finance</u> Smart contract (ETH), AMM (uniswap), perpetual contracts <u>Deep learning for high dimensional problem and stochastic control problems</u> <u>Robo-advising</u>
Teaching/Learning Methodology	The subject will mainly be delivered through lectures and exercise-based tutorials. The teaching and learning approach is mainly problem-solving oriented. The approach aims at the development of financial modeling and quantitative skills and how the techniques can be applied

	to applications in financial technology. Students are encouraged to adopt a deep study approach by employing high level cognitive strategies, such as critical and evaluative thinking, relating, integrating and applying theories to practice.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1.Assignments/Project	20%	✓	✓	✓	✓
	2. Tests	30%	✓	✓	✓	
	3. Examination	50%	✓	✓	✓	✓
	Total	100%				
	Continuous Assessment comprises of assignments/project and online tests. A written examination is held at the end of the semester.					
Student Study Effort Required	Class contact:					
	▪ Lecture			26 Hrs.		
	▪ Tutorial			13 Hrs.		
	Other student study effort:					
	▪ Assignments/Projects			58 Hrs.		
	▪ Self-study			40 Hrs.		
	Total student study effort			137 Hrs.		
Reading List and References	<u>References:</u>					
	D.L.K. Cheun	Handbook of digital currency, Bitcoin, innovation, financial instruments, and big data.			Elsevier, 2015	
	S. Chishti and J. Barberis	The fintech book: the financial technology handbook for investors, entrepreneurs, and visionaries.			John Wiley & Sons, 2016	
	M. King and R. Nesbitt	The technological revolution in financial services			University of Toronto Press, 2020	